21. Percentages

Year 4 · **Supplementary Worksheet** — extra practice & worked breakdowns

Worked Examples

Example 1 — 10% of an amount:

What is 10% of 90?

Answer: 10% means divide by 10: 90 ÷ 10 = **9**.

Example 2 — 30% of an amount:

What is 30% of 50?

Answer: Find 10% first: 50 ÷ 10 = 5. Then multiply by 3 to get 30%: 5 × 3 = **15**.

Example 3 — Building a percentage from 10% and 5%:

What is 15% of 60?

Answer: 10% of 60 = 6. 5% is half of that = 3. 15% = 10% + 5% = 6 + 3 = **9**.

Example 4 — Discount problem:

A £40 jacket has 25% off. What is the new price?

Answer: 25% of £40 = £10 (a quarter of 40). New price: £40 − £10 = **£30**.

Example 5 — Percentage increase:

A plant is 80 cm tall. It grows by 25% of its height. How tall is it now?

Answer: 25% of 80cm = 20cm (a quarter of 80). New height: 80 + 20 = **100 cm**.

Foundation

15 questions

1. What is 50% of 36?

2. What is 25% of 48? A) 9 B) 11 C) 12 D) 15

3. What is 10% of 230?

4. 9 out of 18 is what percentage? Select ALL of the following that are equal to this percentage: A) 50% B) 1/2 C) 0.5 D) 40% E) 9/18 simplified to 1/2

5. What is 75% of 24?

6. What is 50% of 14?

7. What is 25% of 32? A) 6 B) 7 C) 8 D) 9

8. What is 10% of 310?

9. 12 out of 24 is what percentage? Select ALL of the following that are equal to this percentage: A) 50% B) 0.5 C) 1/2 D) 30% E) 12/24 simplified

10. What is 75% of 48?

11. What is 50% of 18?

12. What is 25% of 84? A) 18 B) 21 C) 25 D) 28

13. What is 10% of 70?

14. 15 out of 30 is what percentage?

15. What is 75% of 16?

Standard

20 questions

1. What is 80% of 90? A) 18 B) 72 C) 80 D) 90

2. What is 75% of £12?

3. What is 5% of 180?

4. A £60 jumper has 10% off. Select ALL statements that are TRUE: A) The discount is £6 B) The discount is £10 C) The new price is £54 D) The new price is £50 E) 10% of £60 is £6

5. What is 20% of 180?

6. What is 40% of 85? A) 25 B) 28 C) 34 D) 40

7. What is 75% of £24?

8. What is 5% of 140?

9. A £30 jumper has 10% off. How much is the discount?

10. What is 20% of 320?

11. What is 90% of 60? A) 6 B) 48 C) 54 D) 60

12. What is 75% of £36?

13. What is 5% of 220?

14. A £40 jumper has 20% off. Select ALL statements that are TRUE: A) The discount is £8 B) The discount is £4 C) The new price is £32 D) The new price is £36 E) 20% of £40 is £8

15. What is 20% of 350?

16. What is 90% of 70?

17. What is 75% of £28?

18. What is 5% of 200?

19. A £35 jumper has 10% off. How much is the discount?

20. What is 20% of 410?

Hard

5 questions

1. A plant is 70 cm tall. It grows by 20% of its height. How tall is it now?

2. What is 90% of 250? (Hint: find 10% first, then subtract it from the total.)

3. Mia has £80 and spends 25% of it on a game. Select ALL statements that are TRUE: A) She spends £20 B) She spends £60 C) She has £60 left D) She has £20 left E) 25% of £80 is £20

4. What is 50% of 72, plus 25% of 72?

5. A £90 bike has 10% off in a sale. What is the new price? A) £9 B) £72 C) £81 D) £90

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